



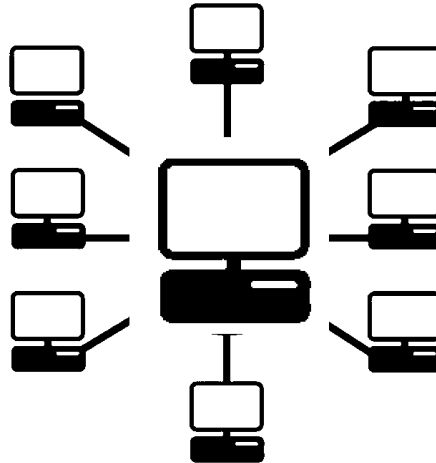
Instructor: Date:

Learner: Regional Branch:

Program: Center:

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**Let's Start!****Grid and Application Architecture****Learning Outcome:**

Exchange written and spoken information on principles, concepts, data base objects related to query analysis, and data base engines. Describe relational database, its concepts, and principles in spoken or written texts.

Can Do Statements:

- I can talk about grid architecture and its advantages.
- I can discuss information about client-server architecture.
- I can talk about multi-tier architecture and its components.

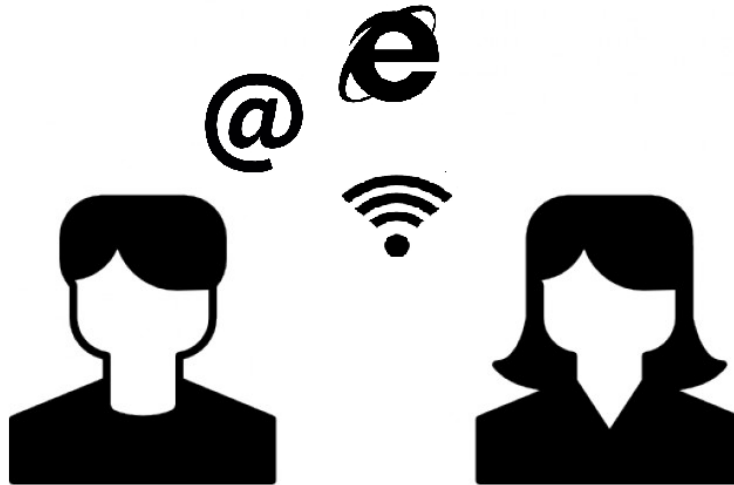
Ask your partner:

- Did the internet change the way companies store information?
- Does a client own the server he uses, why?



Let's Explore!

Grid and application architecture generate a new direction in databases.



Instructions: Read the text below.

Mary: I want to know more about Grid and Application architecture, can you help me?
Billy: Sure! But try to keep up with me, okay? Grid architecture is simple and it can be used by any company. It produces a resilient and lower cost system; this can be seen as one of its main advantages.

Mary: Is it true that independent, modular hardware and software can be connected depending on the needs of the customer with grid computing?

Billy: Yes! The flexibility grid architecture provides is exceptional! Besides, different problems may be solved thanks to it. For example, you won't have application silos that may not be used appropriately, or systems that become difficult to change and expensive to maintain.

Mary: I see. And, what can be said about Application Architecture?

Billy: Well, there are two common database architectures, client/server and multitier. Let me go over the client/server architecture. It's simple; on the one hand, we have a client, which is a database application that requests and processes data; they use different computers from the servers. On the other hand, we have the servers. Servers receive and process the requests that are generated by the client applications. Data must always be managed and kept by a server; the server computers may be optimized to fit the customer needs. This architecture could be surpassed by the multi-tier soon.

Mary: What is multitier, though?

Billy: The multi-tier has the same elements: a client and a database server, but they use one or more application servers as a bridge between the client and the database server. The application servers provide access and remove load work from the end server. The application server is a tool that validates users' credentials, allows a fast connection to the database servers, and even performs the operations requested by the client.



**Let's Practice!**

Why should any company use grid architecture?

Instructions: Circle the correct option to correctly complete each sentence.

1. Even though internet computing has been hanging around less than a century, (they could be said / it could be said / it could be say) that it's not present in all computing environments.
2. The lower cost of the enterprise information (may being considered / may be considered / may be consider) as the biggest advantage of grid architecture.
3. Grid architecture is simple and it (could be used / could been used / could be use) by any company.
4. The fact that independent, modular hardware and software (can connect / can been connected / can be connected) and rejoined according to the needs of the customer is another advantage of grid architecture.
5. Different problems (may be solved / may solving / may be solving) thanks to the flexibility that grid architecture provides.

**Let's Practice!**

What about the multi-tier?

*Instructions: Read the sentences. Write the missing words in the blank. Use: **server could be, can be validated by, can be said about, must be used, be provided.***

Billy: What _____ multi-tier? Well, the multi-tier has the same elements as a client/server architecture: a client and a database server, but one or more application servers _____ as a bridge between the client and the database server. Access must _____ by the application servers, and load work from the end _____ removed, too. Users' credentials _____ the application server. It also allows a fast connection to the database servers, and even performs the operations requested by the client.





Extension Activity

There is always more to learn.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for more information about grid and application architecture.
- Describe the benefits and disadvantages.
- Write a short text explaining each one briefly, using modal verbs in passive voice and phrasal verbs.
- Discuss your work with your partners

Self-Assessment

Yes

No

- I deepen my knowledge on grid and application architecture.
- I describe information through an explanatory paragraph.
- I exchange ideas with my partners.

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Target Words

- **resilient** (adjective): flexible
e.g. It produces a resilient and lower cost system.
- **exceptional** (adjective):
outstanding, excellent qualities
e.g. The flexibility grid architecture provides is exceptional!
- **silo** (noun): it refers to databases
e.g. You won't have application silos that may not be used appropriately.

- **surpass** (verb): to become and do better than others
e.g. This architecture could be surpassed by the multi-tier soon.
- **optimize** (verb): to make something useful and effective
e.g. The server computers may be optimized to fit the customer needs.



Learning Points

Passive of Modal Verbs

Modal verbs are a kind of auxiliary verbs that come with other verbs to express permission, ability or possibility.

These are some verbs with their respective denotation:

Can – May – Might: possibility, ability, permission

Must – Have to: obligation

Should: suggestion

Example:

*Michael **can swim** fast.*

*The athlete **must keep** an appropriate diet.*

Notice that these verbs do not take -es/-s at the end of the verb in the third person singular. For questions, they come at the beginning:

Example:

***Can** you **ride** a bike?*

***Should** I **tell** mom about it?*

We used **modal verbs** in their **passive form** when the important element is the object of the action, and not the person who performs it.

We follow this structure:

Subject + Modal verb + be + Past participle verb + complement

Example:

*Many applications **can be used** to increase the flexibility in our database.*

Examples

- *It **may be surpassed** by the multi-tier soon.*
- *It **could be said** that it's not present in all computing environments.*





Learning Points

Inseparable Phrasal Verbs

Phrasal verbs are verbs composed of two words, a verb and a preposition, that have a different meaning from the original verb. A group of them is separable, while others are inseparable.

Phrasal verbs are made up of two or more words. For example:

Make: create

Up: that points towards or higher

Make + up: to compensate.

Separable phrasal verbs are those verbs whose direct object can be inserted between the verb and the preposition. See this example:

Option 1. I have to **give back** the book to the library.

Option 2. I have to **give** the book **back** to the library.

There are some **phrasal verbs** called "inseparable" because the words can never be set apart.

Examples:

I need to **come out with** a new idea. (**correct**)

I need to **come out** a new idea **with**. (**incorrect**)

I need to **come** a new idea **out with**. (**incorrect**)

Phrasal Verb	Meaning
come out with	produce, utter
go on with	continue
go over	review
settle on	decide
count on	rely
get away with	do something without being caught or punished
pass on	transmit
keep up with	working to be at the same level than someone

Examples

- Internet computing has been hanging around for less than a century.
- Any good company needs to keep up with it.